

Q12 Ans: B

$$I \propto A^2$$

$$\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2} \Rightarrow \sqrt{\frac{0.36}{0.25}} = \frac{A_1}{A_2} \Rightarrow A_1 = 1.2A_2$$

At P (minima), the resultant amplitude, $A_p = 1.2A_2 - A_2 = 0.2 A_2$

$$\frac{I_p}{I_2} = \frac{A_p^2}{A_2^2} \Rightarrow \frac{I_p}{0.25I} = \frac{0.2A_2^2}{A_2^2} = 0.040 \Rightarrow I_p = 0.010I$$

$$\text{Or } A_p = 0.6A - 0.5A = 0.1A$$

$$\Rightarrow I_p = 0.010I$$

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